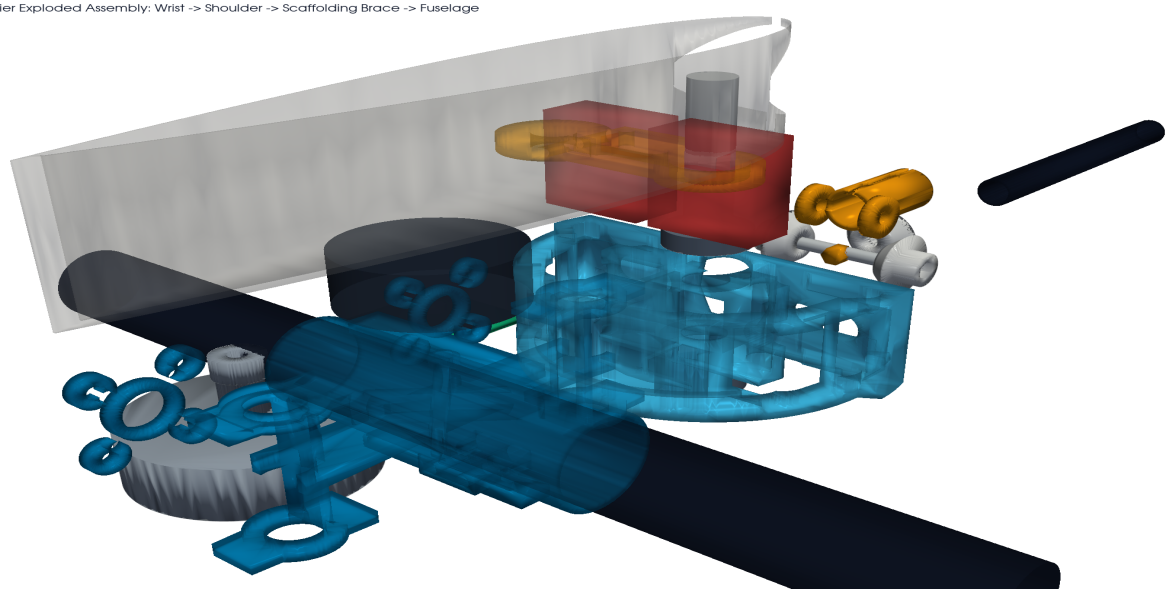
MCO-1D Hierarchical Exploded Installation Guide

2-DOF Differential Bevel Wrist → V10 Flapping Shoulder → Scaffolding Brace → Fuselage Carbon Spine

TIER 1: WRIST	TIER 2: SHOULDER	TIER 3: SCAFFOLDING	FUSELAGE SPINE	TOTAL ASSEMBLY
Bevel Gearset	**Stagecoach Drum**	**Topological Collar**	**Ø20.0mm Carbon**	**73.9g Plastic**
2-DOF Spar Vectoring	*10:1 Capstan (30.5g)*	*Dual Clevis (Y=±35mm)*	*Modular Stations*	*-82% vs V2 Baseline*

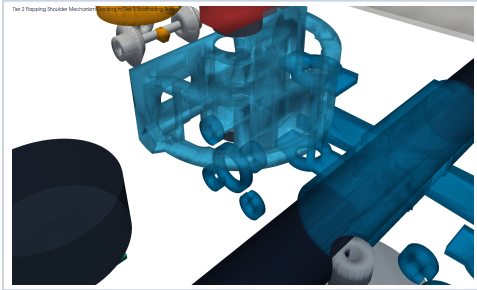
MCO-1D 3-Tier Exploded Assembly: Wrist -> Shoulder -> Scaffolding Brace -> Fuselage



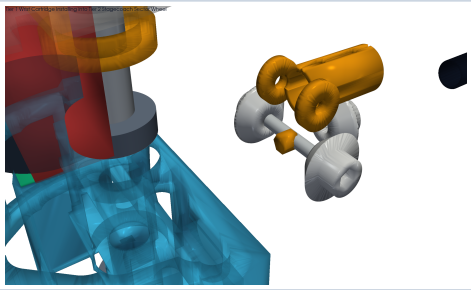
1. Hierarchical Assembly Decomposition & Component Callout Checklist

Tier	ID	Component Name & Mechanical Function	Material & Manufacturing Process	Mass	Assembly Trajectory & Interface
Tier 3 Fuselage & Scaffolding	1 2 3	**Fuselage Carbon Spine Longeron** (Ø20x1.5mm) **Topological Scaffolding Collar** (Bilateral Mount) **Split-Collar M3 Pinch Clamp Fasteners** (Grade 12.9)	Pultruded 3K Carbon Fiber Tube PA12-CF Selective Laser Sintering Alloy Steel Fasteners (Torque 0.90 N·m)	32.0 g 34.5 g 3.2 g	Primary structural datum along X-axis Clamps to spine tube at X=95,275,455,635mm Compresses sleeve with 1,500 N preload
Tier 2 Flapping Shoulder	4 5 6 7 8	**V10 Stagecoach Sector Wheel** (7 Radiating Spokes) **GoolRC 5010 BLDC Motor & Pinion Drum** (10:1) **ODrive Micro Controller & Diametral Magnet** **Dual-Shear Top Brace** (Bearing Tie-Bar) **Ø10mm Titanium Pivot Axle & 2x 6800-2RS Bearings**	PA12-CF (Helical groove, bumper stops) OTS BLDC Motor + Stepped Aluminum Pinion PCB with MPS MA702 IC (1.5mm air gap) PA12-CF Bridge (688-2RS top bearing) Grade 5 Titanium Pin + ABEC-5 Bearings	30.5 g 92.0 g 8.5 g 5.8 g 16.2 g	Seats inside scaffolding clevis (Y=35mm) Bolts to slotted motor stanchion (C=41.25mm) Mounts under motor with 1.50mm air gap Locks top of pinion to Ø10mm main axle Press-fits into clevis lugs & drum hub
Tier 1 Wrist Bevel Cartridge	9 10 11 11 12 13	**Dual AGFRC B24CLM Micro-Servos** (Coreless Titanium) **Differential Gimbal Yoke & Ø4mm Cross-Axle** **3x Quick-Swap 3D-Printed Bevel Gears** (Mod 0.75) **Wing Spar Tube Socket** (Ø10mm Woven Carbon) **Aerodynamic NACA 0018 Airfoil Fairing Shell**	CNC Finned Aluminum Heat-Sink Bodies PA12-CF Yoke + Ground Stainless Axle PA12-CF Quick-Swap Bevel Gearset 3K Twill High-Modulus Carbon Tube Lightweight Foaming PLA (LW-PLA)	42.0 g 7.8 g 4.2 g 18.5 g 9.8 g	Drops into open-air sector wheel bays Inserts into central gimbal cavity (+Y) Meshes inside yoke with zero backlash Bonds into gimbal socket, emerges forward Encloses mechanism with snap-fit latches

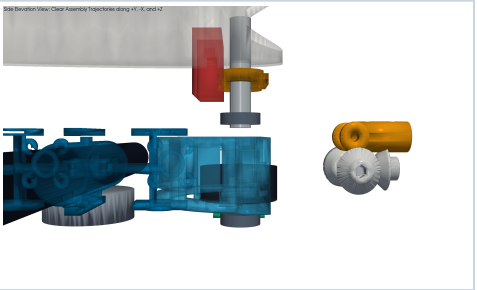
2. Subsystem Installation Trajectories & 3-Tier Assembly Sequence



A. Tier 2 → Tier 3 (Shoulder into Brace)
Stagecoach drum docks into the double-shear clevis (29mm span) at Y=±35mm; Ø10mm pin secures bearings.



B. Tier 1 → Tier 2 (Wrist into Shoulder)
Differential bevel yoke & gears insert through the front trumpet socket into the wheel core; servos drop into side bays.



C. Inline Load Path (Spine → Wing Tip)
Side elevation reveals uninterrupted load trajectories: 220N capstan tension, 4.0g lift shear, and Dyneema wrap.

3. 6-Stage Hierarchical Assembly & Commissioning Protocol

Stage	Subsystem & Mechanical Action	Fastener & Torque Spec	Quality Gate & Critical Verification Metric
ST-01	**Tier 3: Fuselage Spine Collar Clamping** Slide topological scaffolding collar over Ø20.0mm carbon spine tube to designated station (X=95, 275, 455, or 635mm). Tighten split-collar pinch bolts symmetrically.	Fasteners: **2x M3x14mm** Torque: **0.90 N·m** Loctite 243	**Quality Gate**: Verify axial slip resistance (> 1,200 N thrust without movement). Confirm clevis parallelism across Y=±35mm.
ST-02	**Tier 2: Motor, ODrive & Pinion Stanchion Mount** Mount GoolRC 5010 BLDC motor to collar stanchion plate via tensioning slots. Fasten ODrive Micro PCB underneath motor. Press-fit stepped pinion onto shaft.	Motor: **4x M3x10mm (0.85 N·m)** ODrive: **4x M3x6mm (0.45 N·m)** Pinion: **M3 set screw (1.1 N·m)**	**Quality Gate**: Measure magnet-to-MA702 IC air gap using feeler gauge (**1.50 ± 0.15 mm**). Verify pinion axial runout < 0.02mm.
ST-03	**Tier 2: Stagecoach Wheel Clevis Docking** Press dual 6800-2RS bearings into sector drum. Position drum inside scaffolding clevis lugs (Z=-14 to +14mm). Insert Ø10.0mm titanium pivot pin through clevis and drum.	H7/h6 press-fit Bearing lubricant: Klüber Isoflex Topas NB 52	**Quality Gate**: Verify free, zero-play rotation over full ±60° flapping range. Ensure axial float between clevis lugs is 0.2 to 0.4 mm.
ST-04	**Tier 2: Capstan Cable Rigging & Top Brace Tie-Down** Anchor 0.75mm Dyneema SK99 cable into sector drum tensioning bolts. Wrap cable tangentially around stepped pinion. Slide Top Brace over axle and pinion; tighten tie-bolts.	Cable Pre-Tension: **180 N** Brace Bolts: **2x M3x16mm (0.90 N·m)** Axle Clamp: **M3x12mm (0.80 N·m)**	**Quality Gate**: Measure axis-to-axis deflection under 220N peak tension (must be **< 20 µm**). Verify cable stays centered in helical groove.
ST-05	**Tier 1: Wrist Bevel Cartridge Insertion into Drum** Slide differential gimbal yoke into drum front cavity. Insert Ø4mm cross-axle through side gears and spider gear. Drop dual AGFRC B24CLM servos into open cooling bays.	Servo Mount: **4x M2.5x8mm** Torque: **0.40 N·m** Grease: Molykote EM-30L	**Quality Gate**: Check bevel gear backlash (**0.05 to 0.08 mm**). Confirm servos seat flush with aluminum cooling fins 80% exposed to air.
ST-06	**Tier 1: Wing Spar Socket Bonding & Fairing Snap** Insert Ø10.0mm woven carbon wing spar tube into gimbal yoke socket; secure with transverse M2.5 titanium lock-pin. Snap lightweight NACA 0018 fairing over assembly.	Lock Pin: **M2.5 Ti pin** Fairing: **4x snap-latches** Bond: 3M Scotch-Weld DP420	**Quality Gate**: Verify full 2-DOF wrist range of motion (-15° to +36° feathering, ±18° sweep) without fairing contact or cable bind.

CLOSED-LOOP STRUCTURAL INTEGRITY CERTIFICATION:
• **Transverse Tension Arch**: 220N flapping cable pull and 4.0g wing lift couples are reacted across the central scaffolding collar in pure tension, completely shielding the Ø20mm fuselage carbon spine from asymmetric bending fatigue.
• **Rapid Line-Replaceable Unit (LRU)**: The entire Tier 1 wrist differential cartridge can be extracted and swapped in under 90 seconds by removing the Ø4mm cross-axle, without disturbing the Tier 2 capstan rigging or Tier 3 fuselage clamp.